

Carbon Pricing and Industrial Competitiveness

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Abstract

A difference-in-differences design evaluates export performance after the introduction of regional carbon levies. Energy-intensive manufacturers adjust through fuel switching rather than relocation, with modest short-run competitiveness effects.

Keywords: Public Policy; carbon; pricing; industrial; emerging markets; empirical analysis

1. Introduction

A difference-in-differences design evaluates export performance after the introduction of regional carbon levies. Energy-intensive manufacturers adjust through fuel switching rather than relocation, with modest short-run competitiveness effects. This paper contributes to the public policy literature by assembling new evidence and evaluating policy-relevant mechanisms that are often under-studied outside advanced economies.

The remainder of the article is organized as follows. Section 2 reviews related work. Section 3 describes the data and empirical strategy. Section 4 presents results, Section 5 discusses implications, and Section 6 concludes.

2. Literature review

Prior research in public policy emphasizes institutional quality, measurement challenges, and heterogeneous treatment effects across regions. We extend this literature by focusing on settings where administrative records are increasingly available yet still fragmented across agencies.

Our approach complements reduced-form policy evaluations with diagnostics that help distinguish structural adjustment from short-run volatility. In doing so, we respond to recent calls for transparent identification strategies in applied work published by regional economics journals.

3. Data and methods

The analysis combines nationally representative survey waves with firm- and household-level panel components covering 2010–2024. Variables are harmonized across sources, and missing values are addressed using inverse-probability weights where appropriate.

The baseline specification includes region, year, and sector fixed effects, with standard errors clustered at the primary sampling unit. Alternative estimators and placebo tests are reported in the appendix to assess robustness.

4. Results

Baseline estimates indicate economically meaningful effects aligned with the hypotheses stated in the introduction. Magnitudes remain stable after controlling for observable confounders and macroeconomic shocks common to the

sample period.

Event-study coefficients show no significant pre-trends, supporting the interpretation that post-reform changes reflect the policy or market shift under study rather than anticipation effects. Subsample analysis reveals stronger responses among service-sector firms and younger households.

5. Discussion

The findings suggest that design details matter as much as headline policy changes. Complementary investments—in training, digital infrastructure, or enforcement capacity—amplify the gains observed in the primary specification.

E. Schmidt and co-authors note several limitations, including measurement error in informal activity and the possibility of spillovers across neighboring regions that are not fully captured by the current research design.

6. Conclusion

This study documents robust patterns in public policy with clear relevance for policymakers and researchers working on transition economies. The results support targeted reforms paired with monitoring frameworks that make adjustment paths observable to stakeholders.

Future work should extend the sample, incorporate administrative microdata linkages, and evaluate welfare implications using structural models calibrated to local labor and product markets.